Syllabus Details

Syllabus ID:	12223
Syllabus Name:	**Programming Fundamentals_Cơ sở lập trình**
Course Name English:	**Programming Fundamentals**
Subject Code:	**PRF192**
Learning-Teaching Method:	Problem based learning, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	None
Description:	Understand basics computer system and methods of software development, focus on function-oriente Explain basic concepts of programming, function-oriented programming design, modularity, understar Upon completing the course, student should have: 1. Knowledge: (ABET e) - Explain the way to solve a real problem using computer. - Understand the basic concepts computer system, and software development. - Understand the basic concepts of programming, focus on procedure programming, testing and debu 2. Skills in programming: (ABET k) - Read and understand the simple C programs; - Solve real problems using C. 3. Apply learning methods effectively: (ABET i) - Academic reading. - Individual and team work behaviors.
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examinatic - Student is responsible to do all assigned exercises given by instructor in class or at home and submit - Use laptop in class only for learning purpose. - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information.
Tools:	Tools: - Internet - C language utility (ex.DevC++ 6.3)
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1286/QĐ-ĐHFPT dated 11/22/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/22/2024

3 material(s)

No.	Material Description	Author	Publisher
1	Foundations of Programming Using C	Evan Weaver	Creative Commons Attribution-NonCommercial-
2	The C Programming Language	Brian W. Kernighan, Dennis M. Ritchie	Pearson

No.	Material Description	Author	Publisher
3	MOOC: Introduction to C Course	Chris Szalwinsk	Chris Szalwinski and Seneca College.

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand and describe about a program, and the way it works on computer.
2	CLO2	Demonstrate about variables, expressions and basic operations and illustrate them by program examples.
3	CLO3	Explain the meaning and use of Logic constructs in C Language. Describe about programming styles.
4	CLO4	Describe about Modularity and Functions in C programs and their use.
5	CLO5	Understanding and using pointers in programs.
6	CLO6	Discuss about C Libraries and their use.
7	CLO7	Discuss about Arrays, Struct and their use in C programs.
8	CLO8	Discuss about Strings and their use in C programs.
9	CLO9	Discuss about Files and their use in C programs.

View mapping of CLOs to PLOs

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Session	Topic	Learning-Teaching T
1	Introduction and installing tools	Offline
2	Basic concepts computer system; steps to develop a software	Offline
3	Structure of a simple C program; C compiler	Offline
4	Variable and constant used in the C programming language	Offline
5	Understand what is a data type	Offline
6	Basic Memory Operations Implement basic input/output functions using scanf, printf	Offline
7	Expressions: Arithmetic, Relational, Logical, Bit operators, Shorthand Assignment operators	Offline
8	Expressions: Mixing Data Types, Casting, Precedence	Offline
9	Structured Programming: Selection (if-else, switch-case)	Offline
10	Structured Programming: Loop (for, while, do-while)	Offline
11	Review: The lecturer reviews the learned content	Offline
12	Review (contd): The lecturer reviews the learned content	Offline
13	Workshop 1: Practice declaring and initializing values for variables.	Offline
14	Workshop 1: Use expressions to perform simple calculations.	Offline
15	Workshop 1: Practice techniques to control program structure using conditional and loop structures	Offline
16	Modules: What is a module? How to design modules for a program?	Offline
17	Modules: Characteristics of modules	Offline
18	Modules: Principles of defining modules	Offline
19	Functions: Function definitions	Offline
20	Functions: Use a function. Parameters: Pass by value	Offline

Session	Topic	Learning-Teaching T
21	Functions: Differentiate Built-in and User-defined functions	Offline
22	Functions: Extent and Scope of a Variable	Offline
23	Review: The lecturer reviews the learned content	Offline
24	Review (cont): The lecturer reviews the learned content	Offline
25	Workshop 2: Real-world problem analysis techniques for program modularization	Offline
26	Workshop 2:Practice defining and using functions	Offline
27	Workshop 2: Build a program in menu form to call functions	Offline
28	Introduction to Pointers. Where are pointers used?	Offline
29	Pointer Declarations; Pointer operators	Offline
30	Pointers as parameters of a function. Pass by reference	Offline
31	Pointers: Dynamic Allocated Data	Offline
32	Libraries: Standard - stdlib.h; Time - time.h	Offline
33	Libraries (cond): Math - math.h; Character - ctype.h	Offline
34	Input/ Formatted Output	Offline
35	Review: The lecturer reviews the learned content	Offline
36	Progress Test 1: Evaluate students on the theoretical knowledge they have learned	Offline
37	Arrays: Conceps One-Dimensional array and Basic Operations on Array	Offline
38	Matrix: Two-Dimensional Arrays; Inputting, Outputting Data for Matrix and other operations	Offline
39	Arrays: Solve Problems using Arrays (Linear Search and Selection Sort)	Offline
40	Struct: Using struct to organize complex data	Offline
41	Review: The lecturer reviews the learned content	Offline
42	Review: Array or structs	Offline
43	Workshop 3: Practice basic input/ output and operations of array	Offline
44	Workshop 3: Practice with character arrays using library functions in ctype.h	Offline
45	Workshop 3: Manage data with an array of struct	Offline
46	String: Declare/Initialize a string	Offline
47	String: Input/ Output a String using buit-in Functions	Offline
48	Array of strings; operations of String (string.h)	Offline
49	Review: The lecturer reviews the learned content	Offline
50	Review (contd): The lecturer reviews the learned content	Offline
51	Workshop 4: Problem-Solving with String (input/ output/ operations)	Offline
52	File: Basic concepts File	Offline
53	File: File types (Text File, Binary File)	Offline
54	File: Ways for accessing files	Offline
55	File: File Functions	Offline
56	Review: The lecturer reviews the learned content	Offline
57	Workshop 5: Open, read, write, and close files using C file handling functions	Offline
58	Progress Test 2	Offline

Session	Topic	Learning-Teaching T
59	Assigment	Offline
60	Final Review	Offline

6 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	What is C programming language, and what are its primary uses?
2	1	CQ1.2	What are the basic structure and syntax of a C program?
3	1	CQ1.3	What does the #include preprocessor directive do?
4	2	CQ2.1	What is the purpose of the main() function in a C program?
5	2	CQ2.2	How do you comment your code in C?
6	2	CQ2.3	What is the significance of semicolons in C?

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type
1	Assignment	on-going	1	15.0%	>0	at home	CLO2-CLO9	Option 1: a proble
2	Practical Exam	on-going	1	30.0%	>0	85'	CLO2-CLO8	Preferable to be r
3	Progress test	on-going	2	15.0%	>0	20'-40'	CLO1, CLO5, CLO6, CLO9	Option 1: essay or
4	Workshop	on-going	5	10.0%	>0	on class room	CLO1-CLO9	Option 1: practica
5	Final exam	Final exam	1	30.0%	4	60'	CLO1-CLO9	multiple choice